Database Principles: Fundamentals of Design, Implementation, and Management

Lecture 3 & 4:

Entity Relationship Modeling

(Part 1 & 2)

Updated: September 2025

Learning Outcomes

After completing this lecture, students will be able to:

- Describe the main characteristics of entity relationship (ER) components.
- Explain how relationships between entities are defined, refined, and incorporated into the database design process.
- Construct an Entity Relationship Diagram (ERD) based on given business rules or scenarios.

The Entity Relationship Model (ERM)

- ER model forms the basis of an ER diagram.
- ERD represents conceptual database as viewed by end user.
- ERDs depict database's main components:
  - Entities
  - Attributes
  - Relationships

Entities

- Refers to entity set and not to single entity occurrence.
- Corresponds to table and not to row in relational environment.
- In Chen and Crow's Foot models, entity is represented by rectangle with entity's name.
- The entity name, a **noun**, is written in **capital letters.**

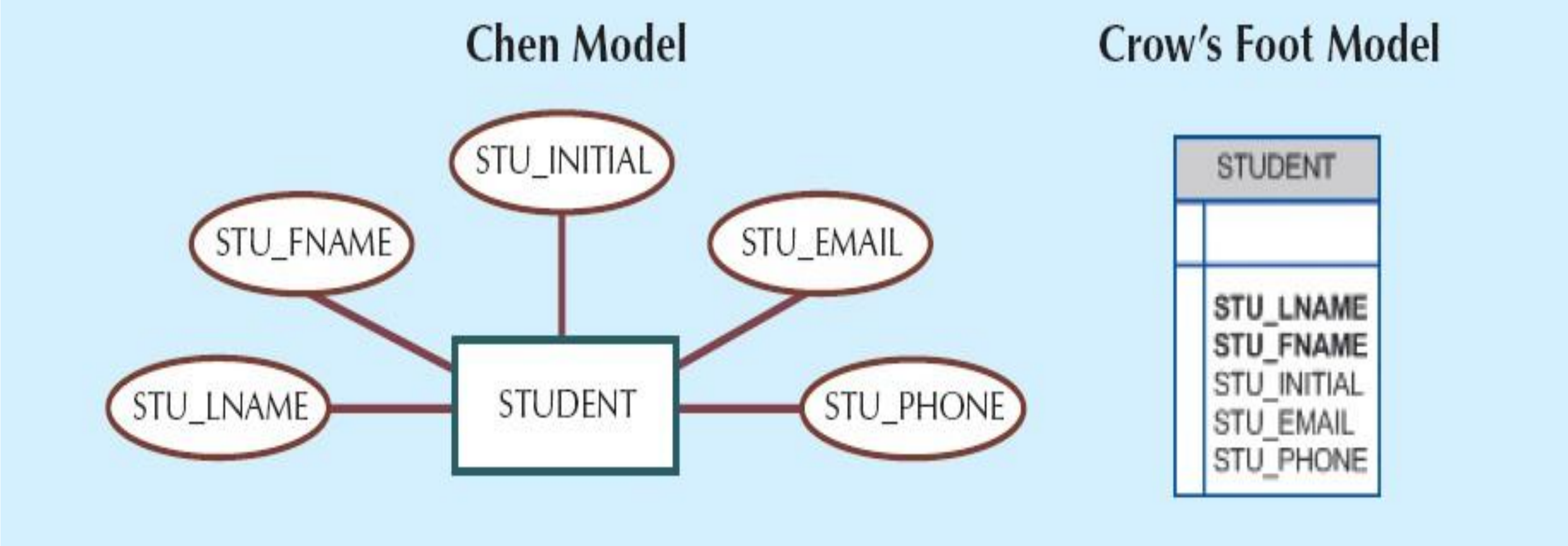
Chen model	**Crow's Foot model**
STUDENT	**STUDENT**

Attributes

- Characteristics (or properties) of entities.
- **Chen notation:** attributes represented by ovals connected to entity rectangle with a line. Each oval contains the name of attribute it represents.
- **Crow's Foot notation:** attributes written in attribute box below entity rectangle.
- Attributes in an Entity-Relationship (ER) model can be grouped into four main categories based on their characteristics:
  - Attribute Optionality
  - Attribute Composition
  - Attribute Multiplicity
  - Attribute Derivation

Attributes (cont'd.)

FIGURE 7.1 The attributes of the STUDENT entity: Chen and Crow's Foot



SOURCE: Course Technology/Cengage Learning

Attributes (cont'd.)

FIGURE 7.2 The CLASS table (entity) components and contents

Database name: Ch07_TinyCollege

CLASS_CODE	CRS_CODE	CLASS_SECTION	CLASS_TIME	ROOM_CODE	PROF_NUM
10012	ACCT-211	1	MWF 8:00-8:50 a.m.	BUS311	105
10013	ACCT-211	2	MWF 9:00-9:50 a.m.	BUS200	105
10014	ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
10015	ACCT-212	1	MWF 10:00-10:50 a.m.	BUS311	301
10016	ACCT-212	2	Th 6:00-8:40 p.m.	BUS252	301
10017	CIS-220	1	MWF 9:00-9:50 a.m.	KLR209	228
10018	CIS-220	2	MWF 9:00-9:50 a.m.	KLR211	114
10019	CIS-220	3	MWF 10:00-10:50 a.m.	KLR209	228
10020	CIS-420	1	W 6:00-8:40 p.m.	KLR209	162
10021	QM-261	1	MWF 8:00-8:50 a.m.	KLR200	114
10022	QM-261	2	TTh 1:00-2:15 p.m.	KLR200	114
10023	QM-362	1	MWF 11:00-11:50 a.m.	KLR200	162
10024	QM-362	2	TTh 2:30-3:45 p.m.	KLR200	162
10025	MATH-243	1	Th 6:00-8:40 p.m.	DRE155	325

SOURCE: Course Technology/Cengage Learning

Attributes (cont'd.)

Attribute Optionality refers to whether an attribute must have a value or may be left empty (NULL) for a particular entity occurrence. It defines if the attribute is required or optional in the database.

1. **Required Attribute**
   - Must have a value; cannot be left empty (NULL).
   - **Example:**
     - StudentID (every student must have an ID)
     - FullName
2. **Optional Attribute**
   - May be left empty (NULL) — not all entity occurrences require a value.
   - **Example:**
     - PhoneNumber (not every student provides it)
     - EmailAddress.

Attributes (cont'd.)

Attribute Composition describes whether an attribute can be divided into smaller, meaningful subparts or not. It helps determine how detailed the data should be stored in a database.

3. **Simple Attribute**
   - Cannot be divided into smaller meaningful parts.
   - **Example:**
     - Age, Gender.
4. **Composite Attribute**
   - Can be divided into smaller subparts, each representing a meaningful component.
   - **Example:**
     - FullName → (FirstName, LastName)
     - Address → (Street, City, Postcode)

Attributes (cont'd.)

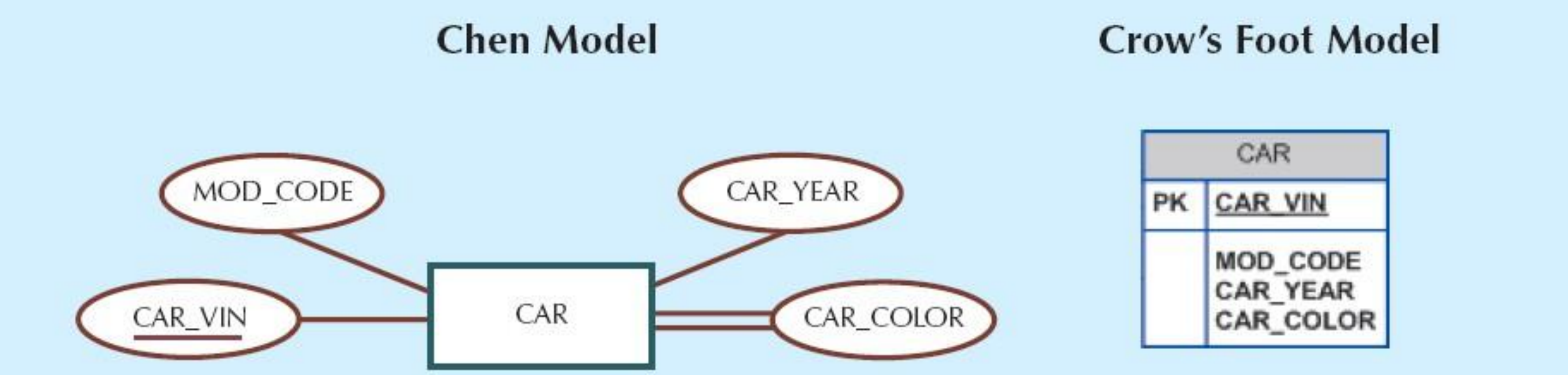
Attribute Multiplicity refers to the number of values an attribute can hold for a single entity occurrence. It determines whether an entity can have only one value or multiple values for the same attribute.

5. **Single-Valued Attribute**
   - Holds only one value for each entity instance.
   - Most attributes in a database are single-valued.
   - **Example:**
     - ICNumber, DateOfBirth, EmployeeName
     - Each person has one IC number and one date of birth.
6. **Multivalued attributes**
   - Can hold more than one value for a single entity instance.
   - **Example:**
     - PhoneNumber (a person may have several numbers)
     - Skill, Qualification

Attributes (cont'd.)

- **Multivalued attributes** should not be implemented:
  - Create several new attributes for each of the original multivalued attributes' components.
  - Create new entity composed of original multivalued attributes' components.

FIGURE 7.3 A multivalued attribute in an entity



SOURCE: Course Technology/Cengage Learning

Attributes (cont'd.)

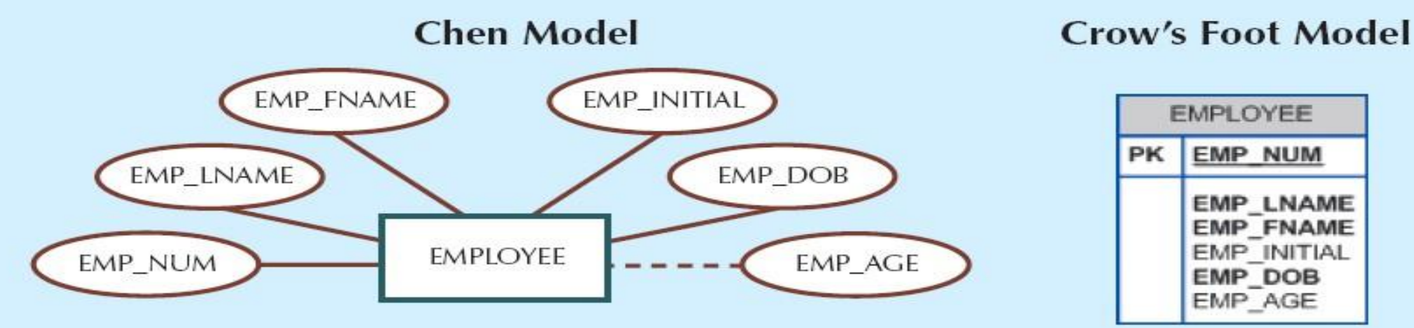
Derived Attribute belongs to its own category, often considered under **Attribute Derivation**.

7. Derived Attribute

- An attribute whose value can be calculated or derived from other attributes in the database.
- Not stored directly in the database; it is computed when needed.
- **Example:**
  - Age derived from DateOfBirth
  - TotalPrice derived from (Quantity × UnitPrice)

Attributes (cont'd.)

FIGURE 7.6 Depiction of a derived attribute



SOURCE: Course Technology/Cengage Learning

TABLE 7.2 Advantages and Disadvantages of Storing Derived Attributes

	DERIVED ATTRIBUTE	
	STORED	NOT STORED
Advantage	Saves CPU processing cycles Saves data access time Data value is readily available Can be used to keep track of historical data	Saves storage space Computation always yields current value
Disadvantage	Requires constant maintenance to ensure derived value is current, especially if any values used in the calculation change	Uses CPU processing cycles Increases data access time Adds coding complexity to queries

Attributes (cont'd.)

- **Attribute Domain**:
  - Specifies the type, range, or set of values that the attribute can hold.
  - Ensures data integrity and validity.
  - Helps prevent invalid or inconsistent entries.
  - Attributes may share a domain.
  - Domains are usually documented in the data dictionary.
  - **Examples:**
    - Numeric: Integer, Decimal, e.g., Age = 0–120
    - Character/String: Letters, Codes, e.g., FirstName = A–Z characters
    - Date/Time: Specific date format, e.g., BirthDate = yyyy-mm-dd
    - numerated Values: Predefined set, e.g., Gender = {Male, Female}, Status = {Active, Inactive}

Attributes (cont'd.)

- **Identifiers**: one or more attributes that uniquely identify each entity instance
  - Is an attribute (or a set of attributes) that uniquely identifies each instance of an entity.
  - Also called a primary key.
  - Identifiers are crucial for ensuring entity integrity.
  - Types of Identifiers: Simple Identifier and Composite Identifier
- **Simple Identifier:**
  - A single attribute that uniquely identifies an entity.
  - **Example:** StudentID, ICNumber
- **Composite Identifier:**
  - A combination of two or more attributes used together to uniquely identify an entity.
  - **Example:** (OrderID, ProductID) in Order Detail

Relationships

- Association between entities.
- Participants are entities that participate in a relationship.
- Relationships between entities always operate in both directions.
- Relationship can be classified as:
  - **One-to-many (1:M) relationship**
  - **One-to-one (1:1) relationship**
  - **Many-to-many (M:N or M:M) relationship**
- In a Crow's Foot notation, relationships are represented by lines connecting entities.
- Relationship classification is difficult to establish if only one side of the relationship is known.

Connectivity and Cardinality

- **Connectivity**
  - Describes the relationship classification.
- **Cardinality**
  - Expresses minimum and maximum number of entity occurrences associated with one occurrence of related entity.
- Established by very concise statements known as business rules.

Connectivity and Cardinality (cont'd.)

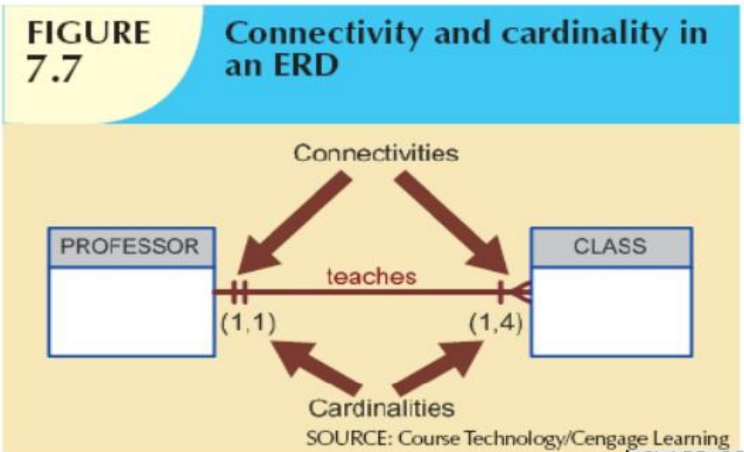


FIGURE 7.7 Connectivity and cardinality in an ERD

SOURCE: Course Technology/Cengage Learning

Connectivity - 1:M
Business Rule :
Each professor teaches up to four classes.
Each class is taught by one and only one professor.

PROF_NUM	PROF_NAME	DEPT_ID
105	Ali	ICA
114	Zulaiha	MI
162	Bakri	SE
228	Aisyah	SKK

CLASS_CODE	CRS_CODE	CLASS_SECTION	CLASS_TIME	ROOM_CODE	PROF_NUM
10012	ACCT-211	1	MWF 8:00-8:50 a.m.	BUS311	105
10013	ACCT-211	2	MWF 9:00-9:50 a.m.	BUS200	105
10014	ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
10015	ACCT-212	1	MWF 10:00-10:50 a.m.	BUS311	301
10016	ACCT-212	2	Th 6:00-8:40 p.m.	BUS252	301
10017	CIS-220	1	MWF 9:00-9:50 a.m.	KLR209	228
10018	CIS-220	2	MWF 9:00-9:50 a.m.	KLR211	114
10019	CIS-220	3	MWF 10:00-10:50 a.m.	KLR209	228
10020	CIS-420	1	W 6:00-8:40 p.m.	KLR209	162
10021	QM-261	1	MWF 8:00-8:50 a.m.	KLR200	114
10022	QM-261	2	TTh 1:00-2:15 p.m.	KLR200	114
10023	QM-362	1	MWF 11:00-11:50 a.m.	KLR200	162
10024	QM-362	2	TTh 2:30-3:45 p.m.	KLR200	162
10025	MATH-243	1	Th 6:00-8:40 p.m.	DRE155	325

The professor's primary key value occurs at least once and no more than four times as foreign key values in the class table

Connectivity (cont'd.)

1. 1:M relationship

- Relational modeling ideal.
- Should be the norm in any relational database design.

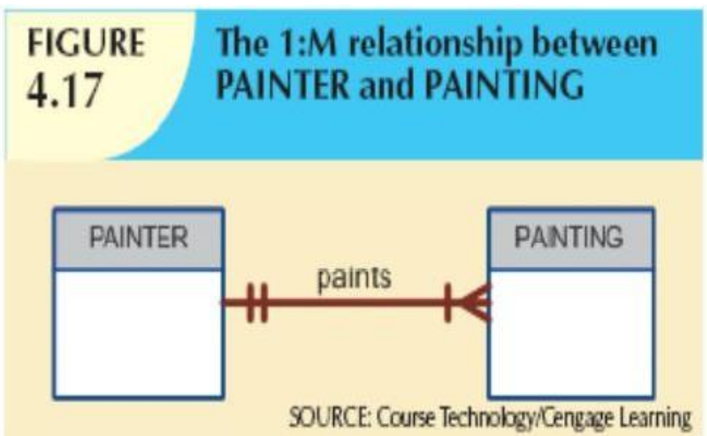
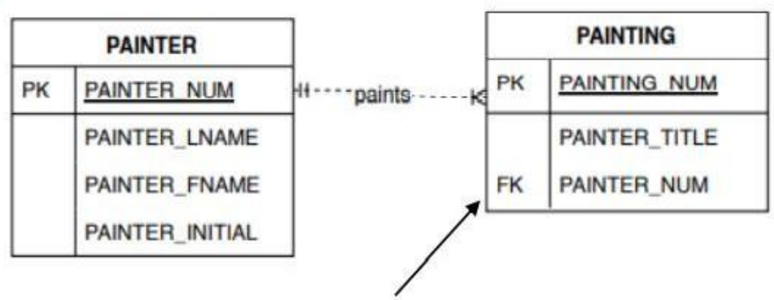


FIGURE 4.17 The 1:M relationship between PAINTER and PAINTING



Foreign Key – include at M side

Business Rules:
Each painter paints one or many paintings
Each painting is painted by one and only one painter.

Connectivity (cont'd.)

FIGURE 4.18 The implemented 1:M relationship between PAINTER and PAINTING

Database name: Ch04_Museum

Table name: PAINTER
Primary key: PAINTER_NUM
Foreign key: none

PAINTER_NUM	PAINTER_LNAME	PAINTER_FNAME	PAINTER_INITIAL
123	Ross	Georgette	P
126	Itero	Julio	G

Table name: PAINTING
Primary key: PAINTING_NUM
Foreign key: PAINTER_NUM

PAINTING_NUM	PAINTING_TITLE	PAINTER_NUM
1338	Dawn Thunder	123
1339	Vanilla Roses To Nowhere	123
1340	Tired Flounders	126
1341	Hasty Exit	123
1342	Plastic Paradise	126

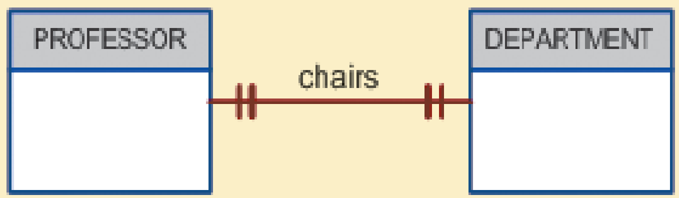
SOURCE: Course Technology/Cengage Learning

Connectivity (cont'd.)

2. **1:1 relationship**
   - Should be rare in any relational database design.
   - One entity related to only one other entity, and vice versa.
   - Sometimes means that entity components were not defined properly.
   - Could indicate that two entities actually belong in the same table.
   - Certain conditions absolutely require their use.

Connectivity (cont'd.)

FIGURE 4.21 The 1:1 relationship between PROFESSOR and DEPARTMENT



SOURCE: Course Technology/Cengage Learning

Business Rules:
A professor chairs one and only one department.
A department is chaired by one and only one professor.

PROF_NUM	PROF_FNAME
105	Ali
114	Zulaiha
162	Bakri
228	Aisyah
301	Amira
342	Naufal

DEPT_ID	DEPT_NAME	HOD
SE	Software Engineering	162
MI	Interactive Media	114
SKK	Computer and Communication System	228
ICA	Intelligent Computing and Analytics	301

Connectivity (cont'd.)

Which side to put FK for 1:1 relationship?

PROF_NUM	PROF_NAME	DEPT_ID_CHAIR
105	Ali	
114	Zulaiha	MI
162	Bakri	SE
228	Aisyah	SKK
301	Amira	ICA
342	Naufal	
152	Azman	
106	Rahim	
107	Raina	
108	Sufiah	
109	Hariz	
110	Nurin	

DEPT_ID	DEPT_NAME
SE	Software Engineering
MI	Interactive Media
SKK	Computer and Communication System
ICA	Intelligent Computing and Analytics

Many null values

Connectivity (cont'd.)

3. M:N relationships

– Cannot be implemented as such in the relational model.
– M:N relationships can be changed into 1:M relationships.
– Implemented by breaking it up to produce a set of 1:M relationships.
– Avoid problems inherent to M:N relationship by creating a **composite (bridge/associative) entity**.
– Includes as foreign keys the primary keys of tables to be linked.

Connectivity (cont'd.)

FIGURE 4.23 The ERM's M:N relationship between STUDENT and CLASS

SOURCE: Course Technology/Cengage Learning

Business Rules:
A class has one or many students.
A student attends one or many classes.

How to implement the M:N relationship between STUDENT and CLASS?

Table name: STUDENT
Primary key: STU_NUM
Foreign key: none

STU_NUM	STU_LNAME
321452	Bowser
324257	Smithson

Table name: CLASS
Primary key: CLASS_CODE
Foreign key: CRS_CODE

CLASS_CODE	CRS_CODE	CLASS_SECTION	CLASS_TIME	CLASS_ROOM	PROF_NUM
10014	ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
10018	CIS-220	2	MWF 9:00-9:50 a.m.	KLR211	114
10021	QM-261	1	MWF 8:00-8:50 a.m.	KLR200	114

Connectivity (cont'd.)

FIGURE 4.25 Converting the M:N relationship into two 1:M relationships

Database name: Ch04_CollegeTry2

Table name: STUDENT
Primary key: STU_NUM
Foreign key: none

STU_NUM	STU_LNAME
321452	Bowser
324257	Smithson

Table name: ENROLL
Primary key: CLASS_CODE + STU_NUM
Foreign key: CLASS_CODE, STU_NUM

CLASS_CODE	STU_NUM	ENROLL_GRADE
10014	321452	C
10014	324257	B
10018	321452	A
10018	324257	B
10021	321452	C
10021	324257	C

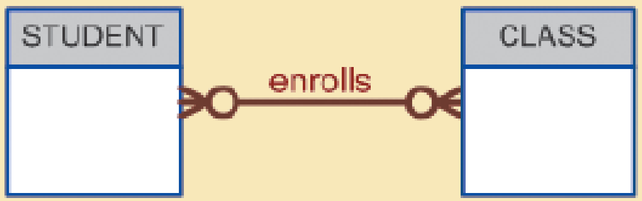
Table name: CLASS
Primary key: CLASS_CODE
Foreign key: CRS_CODE

CLASS_CODE	CRS_CODE	CLASS_SECTION	CLASS_TIME	CLASS_ROOM	PROF_NUM
10014	ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
10018	CIS-220	2	MWF 9:00-9:50 a.m.	KLR211	114
10021	QM-261	1	MWF 8:00-8:50 a.m.	KLR200	114

SOURCE: Course Technology/Cengage Learning

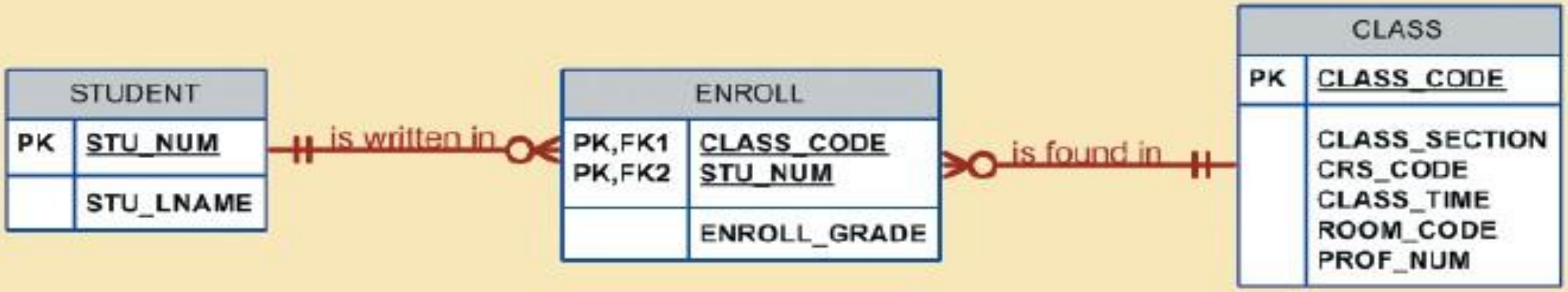
Connectivity (cont'd.)

FIGURE 7.24 The M:N relationship between STUDENT and CLASS



SOURCE: Course Technology/Cengage Learning

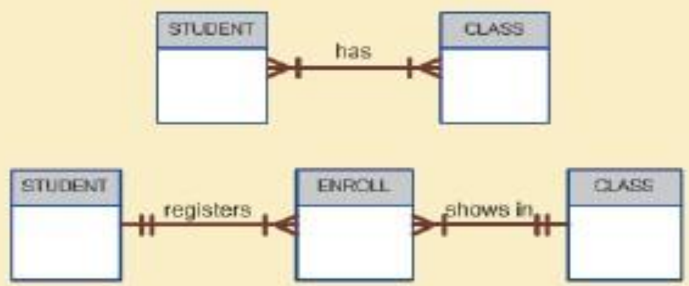
FIGURE 7.25 A composite entity in an ERD



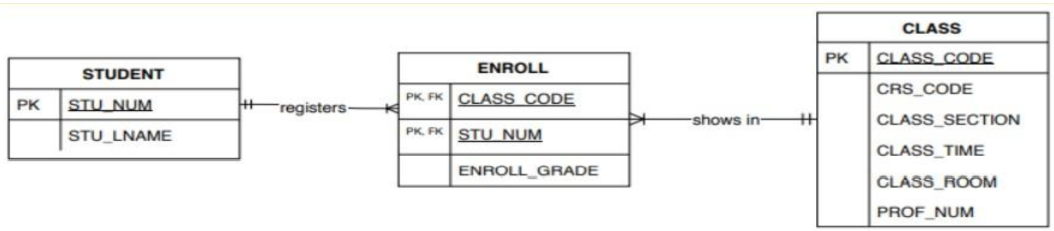
SOURCE: Course Technology/Cengage Learning

Connectivity (cont'd.)

FIGURE 4.26 Changing the M:N relationship to two 1:M relationships



SOURCE: Course Technology/Cengage Learning

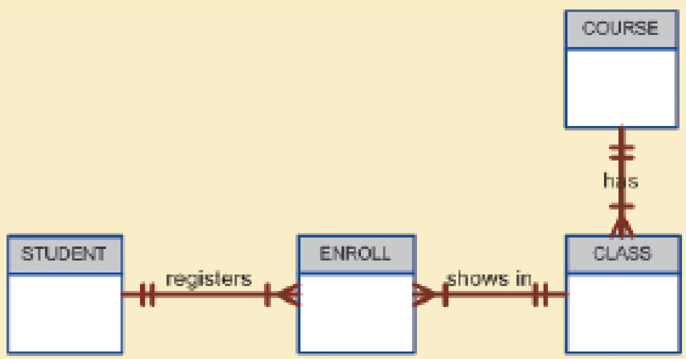


Business Rules:
A student registers one or many classes in enrolment.
An enrollment is made by one and only one student.

A class shows in one or many enrolments.
An enrolment has one and only one class.

Connectivity (cont'd.)

FIGURE 4.27 The expanded entity relationship model



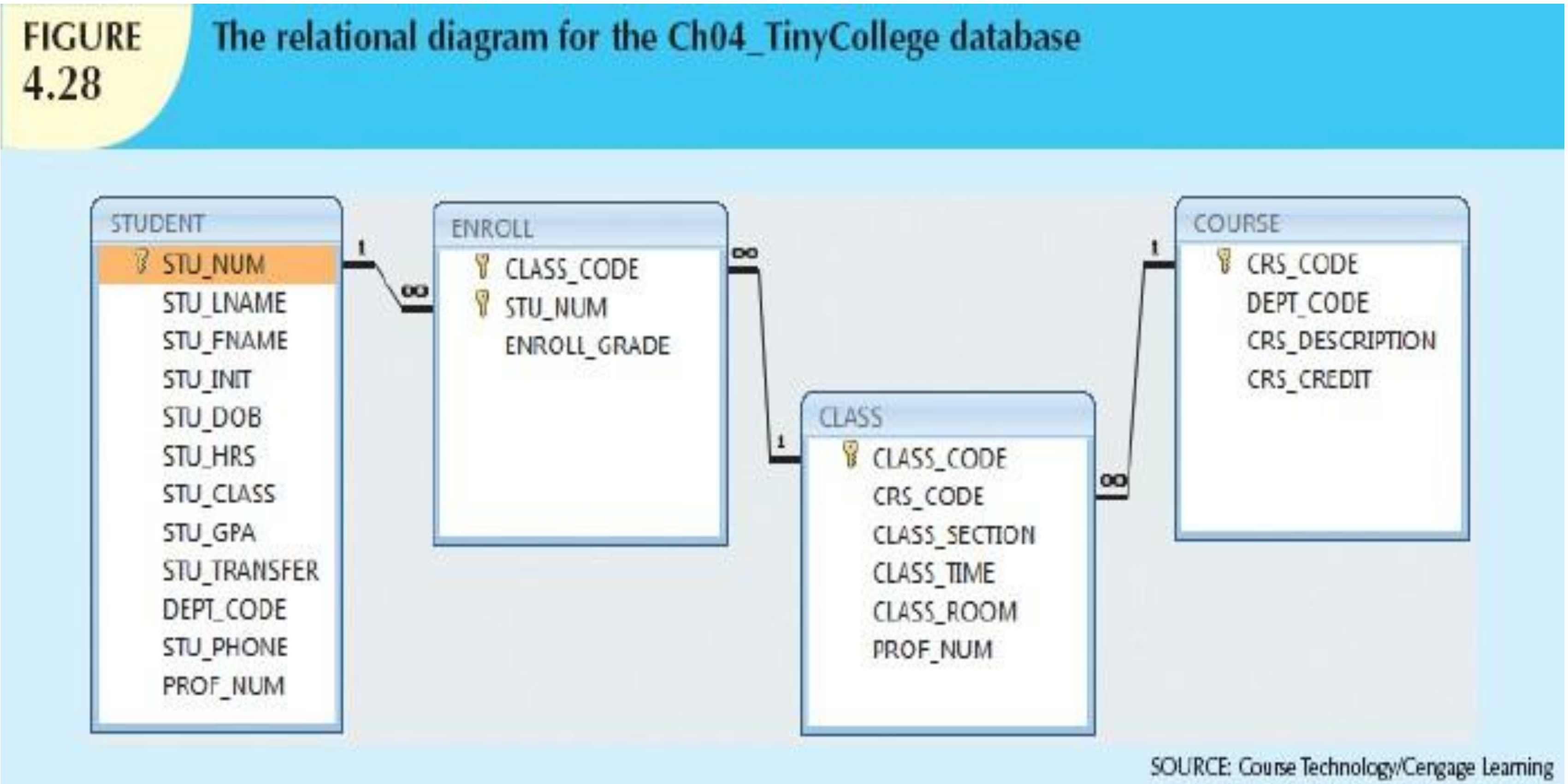
SOURCE: Course Technology/Cengage Learning

Business Rules:
A student registers one or many classes in enrolment.
An enrollment is made by one and only one student.

A class shows in one or many enrolments.
An enrolment has one and only one class.

A course has one or many classes.
A class belongs to one and only one course.

Connectivity (cont'd.)



Existence Dependence

1. Existence dependence

- Existence Dependence (also called **Weak Entity**) refers to a situation where one entity cannot exist without being associated with another entity.
- In other words, the existence of a child (or dependent) entity depends on the existence of a parent (or owner) entity.
- **Example:**
  - Dependent entity: Order Detail
  - Parent entity: Order

  An Order Detail record cannot exist without a corresponding Order.
- **In ERD (Crow's Foot Notation):**
  - This relationship is represented as a mandatory relationship, where the minimum cardinality is 1.
  - Example: An Order Detail must be linked to exactly one Order.

Existence Dependence (cont'd.)

2. Existence independence

- Existence Independence (also called **Strong or Regular Entity**) refers to a situation where an entity can exist independently of any other entity in the database.
- In other words, it does not depend on the existence of another entity for its identification or survival.
- **Example:**
  - Independent entity: Customer
  - A Customer can exist in the database even if they have not placed any orders.
- **In ERD (Crow's Foot Notation):**
  - It participates in relationships optionally, meaning its minimum cardinality is 0 (it can exist without the related entity).

Relationship Strength

- **Relationship Strength** refers to how strongly an entity is dependent on another entity for its existence within a database model.
- In the Entity-Relationship (ER) Model, relationship strength determines whether a relationship is strong or weak.

Relationship Strength (cont'd.)

1. **Weak (non-identifying) relationships**
   - Exists when the child entity is existence-independent of the parent entity.
   - Exists if PK of related entity does not contain PK component of parent entity
   - The entities can exist separately.
   - Represented in Crow's Foot ERD by a **dashed line**.
   - **Example:**
     - Customer and Order — a Customer may exist without placing any Order.

Relationship Strength (cont'd.)

FIGURE 7.8 A weak (non-identifying) relationship between COURSE and CLASS

Table name: COURSE

Database name: Ch07_TinyCollege

CRS_CODE	DEPT_CODE	CRS_DESCRPTION	CRS_CREDIT
ACCT-211	ACCT	Accounting I	3
ACCT-212	ACCT	Accounting II	3
CIS-220	CIS	Intro. to Microcomputing	3
CIS-420	CIS	Database Design and Implementation	4
MATH-243	MATH	Mathematics for Managers	3
QM-261	CIS	Intro. to Statistics	3
QM-362	CIS	Statistical Applications	4

MATH-243 does not have any classes

Table name: CLASS

CLASS_CODE	CRS_CODE	CLASS_SECTION	CLASS_TIME	ROOM_CODE	PROF_NUM
10012	ACCT-211	1	MWF 8:00-8:50 a.m.	BUS311	105
10013	ACCT-211	2	MWF 9:00-9:50 a.m.	BUS200	105
10014	ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
10015	ACCT-212	1	MWF 10:00-10:50 a.m.	BUS311	301
10016	ACCT-212	2	Th 6:00-8:40 p.m.	BUS252	301
10017	CIS-220	1	MWF 9:00-9:50 a.m.	KLR209	228
10018	CIS-220	2	MWF 9:00-9:50 a.m.	KLR211	114
10019	CIS-220	3	MWF 10:00-10:50 a.m.	KLR209	228
10020	CIS-420	1	W 6:00-8:40 p.m.	KLR209	162
10021	QM-261	1	MWF 8:00-8:50 a.m.	KLR200	114
10022	QM-261	2	TTh 1:00-2:15 p.m.	KLR200	114
10023	QM-362	1	MWF 11:00-11:50 a.m.	KLR200	162
10024	QM-362	2	TTh 2:30-3:45 p.m.	KLR200	162

SOURCE: Course Technology/Cengage Learning

Relationship Strength (cont'd.)

2. **Strong (identifying) relationships**
   - Exists when the child entity is existence-dependent on the parent entity and has a primary key that includes the parent entity's primary key.
   - Usually occurs between a strong (independent) and a weak (dependent) entity.
   - Represented in Crow's Foot ERD by a **solid line**.
   - **Example:**
     - Order (parent) and Order Detail (child)
     - Order Detail cannot exist without an Order.

Relationship Strength (cont'd.)

FIGURE 7.9 A strong (identifying) relationship between COURSE and CLASS

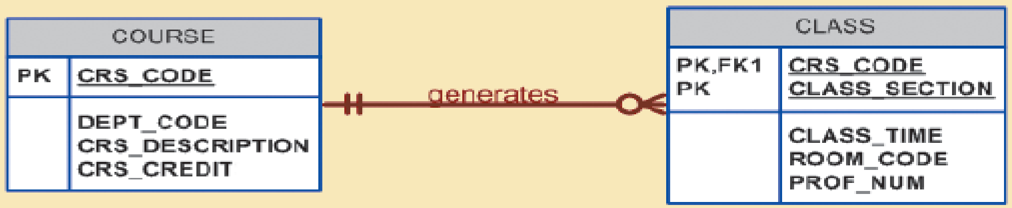


Table name: COURSE

Database name: Ch07_TinyCollege_Alt

CRS_CODE	DEPT_CODE	CRS_DESCRIPTION	CRS_CREDIT
ACCT-211	ACCT	Accounting I	3
ACCT-212	ACCT	Accounting II	3
CIS-220	CIS	Intro. to Microcomputing	3
CIS-420	CIS	Database Design and Implementation	4
MATH-243	MATH	Mathematics for Managers	3
QM-261	CIS	Intro. to Statistics	3
QM-362	CIS	Statistical Applications	4

Table name: CLASS

CRS_CODE	CLASS_SECTION	CLASS_TIME	ROOM_CODE	PROF_NUM
ACCT-211	1	MWF 8:00-8:50 a.m.	BUS311	105
ACCT-211	2	MWF 9:00-9:50 a.m.	BUS200	105
ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
ACCT-212	1	MWF 10:00-10:50 a.m.	BUS311	301
ACCT-212	2	Th 6:00-8:40 p.m.	BUS252	301
CIS-220	1	MWF 9:00-9:50 a.m.	KLR209	228
CIS-220	2	MWF 9:00-9:50 a.m.	KLR211	114
CIS-220	3	MWF 10:00-10:50 a.m.	KLR209	228
CIS-420	1	W 6:00-8:40 p.m.	KLR209	162
MATH-243	1	Th 6:00-8:40 p.m.	DRE155	325
QM-261	1	MWF 8:00-8:50 a.m.	KLR200	114
QM-261	2	TTh 1:00-2:15 p.m.	KLR200	114
QM-362	1	MWF 11:00-11:50 a.m.	KLR200	162
QM-362	2	TTh 2:30-3:45 p.m.	KLR200	162

SOURCE: Course Technology/Cengage Learning

Relationship Participation

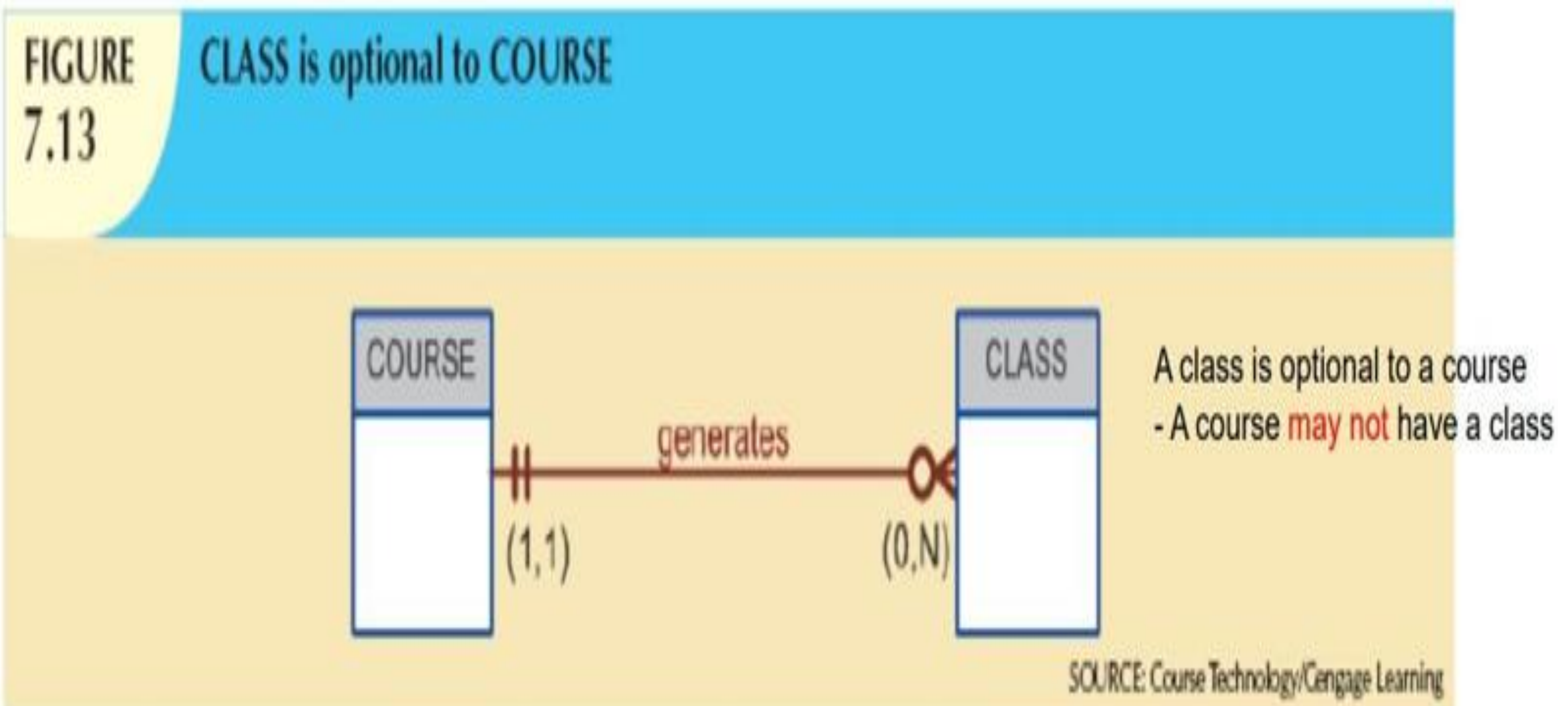
- **Relationship Participation** describes whether all or only some entity occurrences are involved in a relationship.
- It indicates the degree of involvement of an entity in a relationship within an ER model.
- Types of relationship participation:
  - Optional participation
  - Mandatory participation

Relationship Participation (cont'd.)

1. Optional participation

- One entity occurrence does not require corresponding entity occurrence in particular relationship.
- An entity does not have to participate in the relationship.
- The minimum cardinality is 0.
- Represented in Crow's Foot notation by a **circle (O)** near the entity.
- **Example:**
  - A Customer may place an Order, but it's possible that some customers have not placed any orders yet.

Relationship Participation (cont'd.)



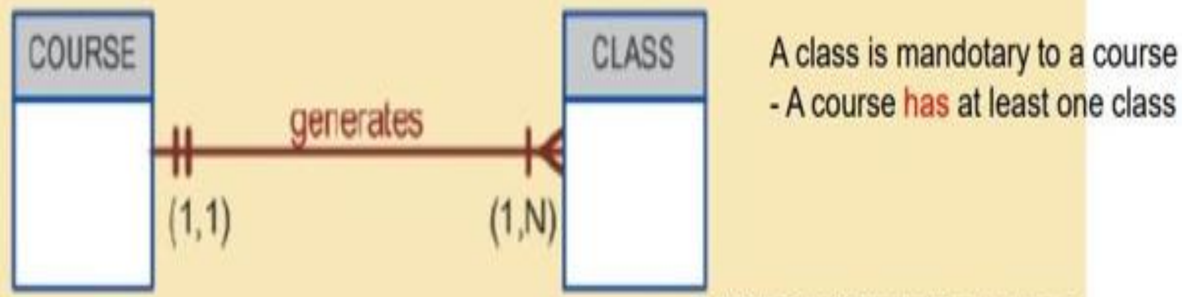
Relationship Participation (cont'd.)

2. Mandatory participation

- One entity occurrence requires corresponding entity occurrence in particular relationship.
- An entity must participate in the relationship.
- The minimum cardinality is 1.
- Represented in Crow's Foot notation by a **short perpendicular line (|)** near the entity.
- **Example:**
  - An Order Detail must be linked to an Order — it cannot exist without one.

Relationship Participation (cont'd.)

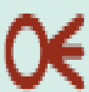
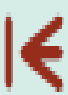
FIGURE 7.14 COURSE and CLASS in a mandatory relationship



SOURCE: Course Technology/Cengage Learning

Relationship Participation (cont'd.)

TABLE 7.3 Crow's Foot Symbols

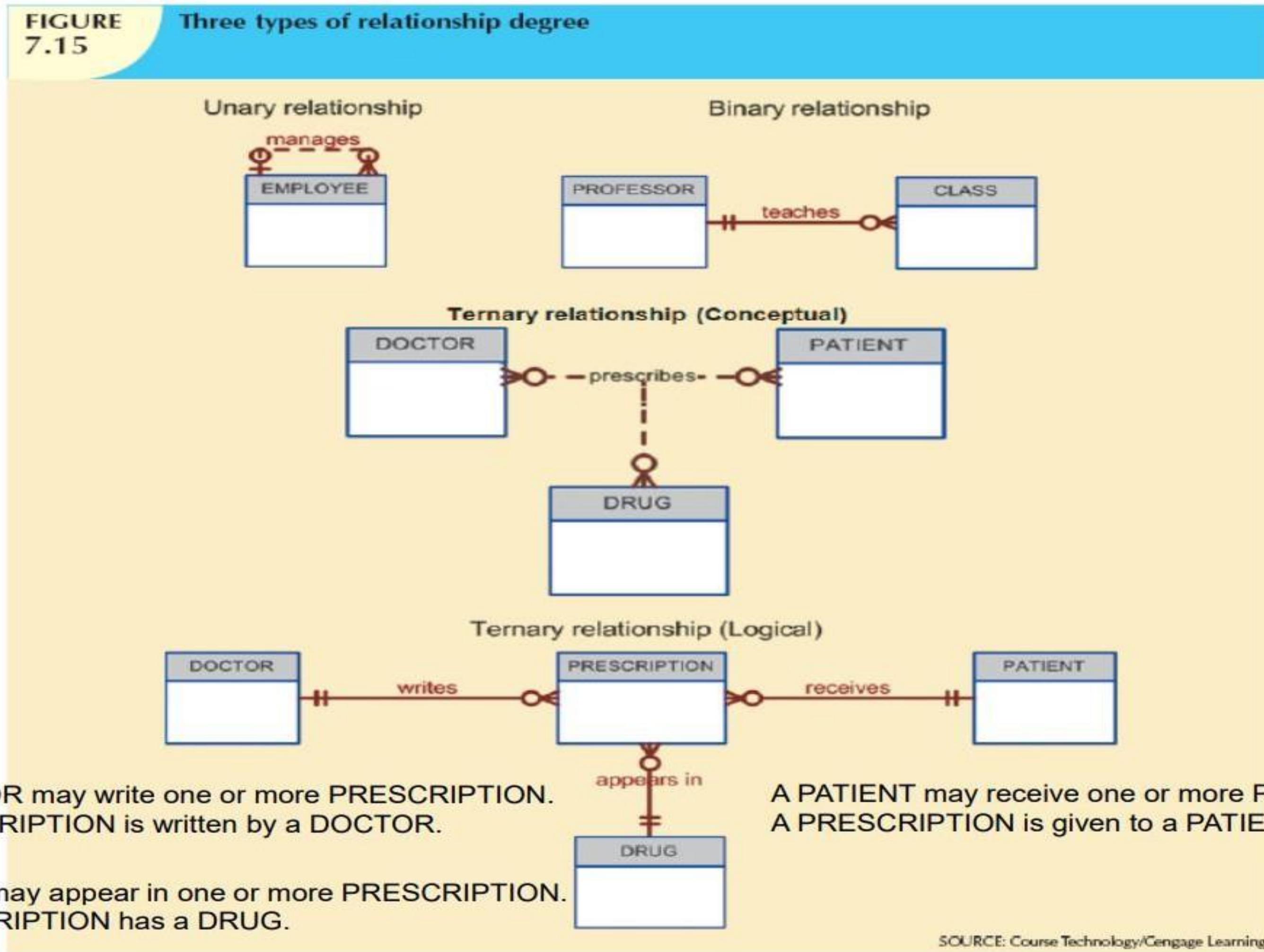
CROW'S FOOT SYMBOLS	CARDINALITY	COMMENT
	(0,N)	Zero or many; the "many" side is optional.
	(1,N)	One or many; the "many" side is mandatory.
	(1,1)	One and only one; the "1" side is mandatory.
	(0,1)	Zero or one; the "1" side is optional.

Relationship Degree

- Relationship Degree refers to the number of entities that participate in a relationship within an ER model.

- **Types of Relationship Degree:**
  - **Unary relationship**
  - **Binary relationship**
  - **Ternary relationship**

Relationship Degree (cont'd.)



A DOCTOR may write one or more PRESCRIPTION.
A PRESCRIPTION is written by a DOCTOR.

A PATIENT may receive one or more PRESCRIPTION.
A PRESCRIPTION is given to a PATIENT.

A DRUG may appear in one or more PRESCRIPTION.
A PRESCRIPTION has a DRUG.

Relationship Degree (cont'd.)

1. **Unary relationship (Degree 1)**
   - Association is maintained within single entity
   - Also called a **recursive relationship** — an entity is related to itself.
   - **Example:**
     - Employee manages another Employee.
     - Part may be a component of another Part.

Relationship Degree (cont'd.)

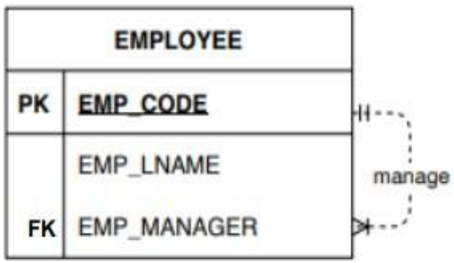
FIGURE 7.22 Implementation of the 1:M recursive relationship "EMPLOYEE manages EMPLOYEE"

Database name: Ch07_PartCo

Table name: EMPLOYEE_V2

EMP_CODE	EMP_LNAME	EMP_MANAGER
101	Waddell	102
102	Orincona	
103	Jones	102
104	Reballoh	102
105	Robertson	102
106	Deltona	102

SOURCE: Course Technology/Cengage Learning



Relationship Degree (cont'd.)

FIGURE 7.19 Another unary relationship: "PART contains PART"

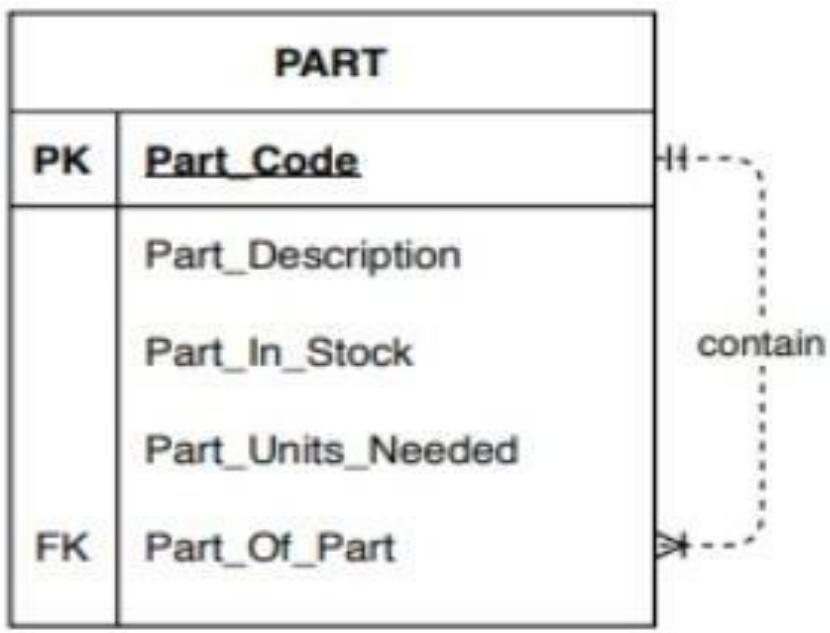
Table name: PART_V1

Database name: CH07_PartCo

PART_CODE	PART_DESCRIPTION	PART_IN_STOCK	PART_UNITS_NEEDED	PART_OF_PART
AA21-6	2.5 cm. washer, 1.0 mm. rin	432	4	C-130
AB-121	Cotter pin, copper	1034	2	C-130
C-130	Rotor assembly	36		
E129	2.5 cm. steel shank	128	1	C-130
X10	10.25 cm. rotor blade	345	4	C-130
X34AW	2.5 cm. hex nut	879	2	C-130

SOURCE: Course Technology/Cengage Learning

Relationship Degree (cont'd.)



Relationship Degree (cont'd.)

FIGURE 7.20
Implementation of the M:N recursive relationship "PART contains PART"

Table name: COMPONENT

Database name: Ch07_PartCo

COMP_CODE	PART_CODE	COMP_PARTS_NEEDED
C-130	AA21-6	4
C-130	AB-121	2
C-130	E129	1
C-131A2	E129	1
C-130	X10	4
C-131A2	X10	1
C-130	X34AW	2
C-131A2	X34AW	2

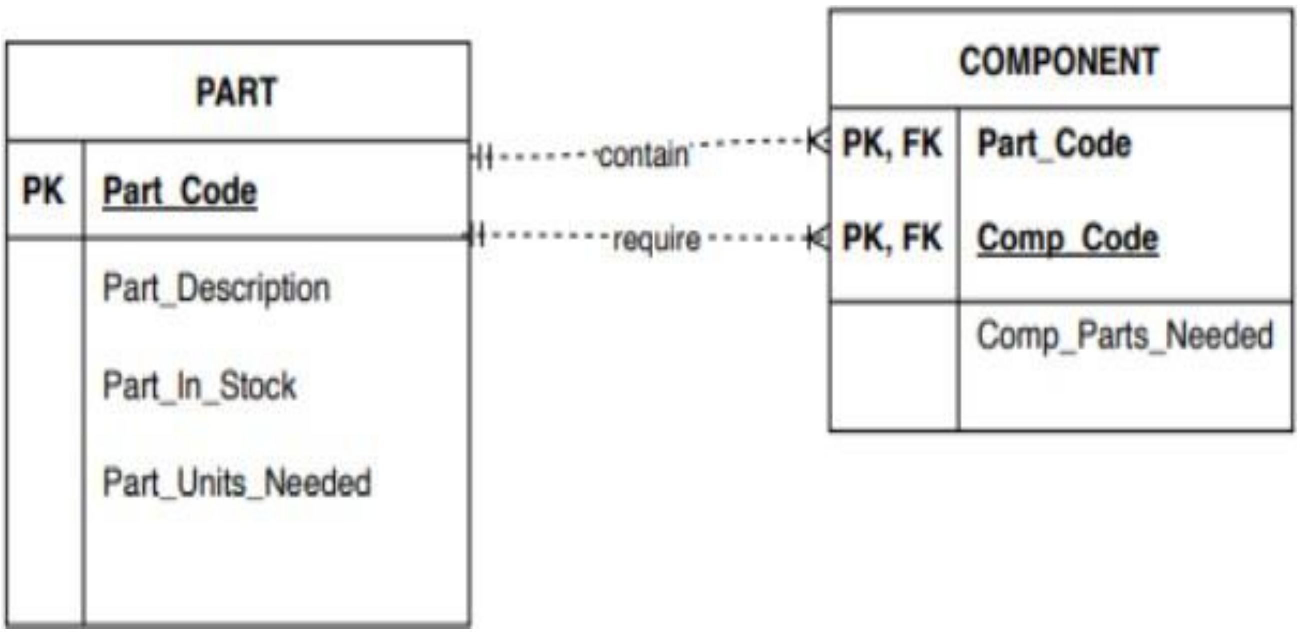
Table name: PART

PART_CODE	PART_DESCRIPTION	PART_IN_STOCK
AA21-6	2.5 cm. washer, 1.0 mm. rim	432
AB-121	Cotter pin, copper	1034
C-130	Rotor assembly	36
E129	2.5 cm. steel shank	128
X10	10.25 cm. rotor blade	345
X34AW	2.5 cm. hex nut	879

SOURCE: Course Technology/Cengage Learning

Rotor Assembly(C-130) is composed of AA21-6, AB-121, E129,X10, X34AW
X10 is part of C-13-, C-131A2

Relationship Degree (cont'd.)



Relationship Degree (cont'd.)

2. **Binary relationship (Degree 2)**
   - Two entities are associated.
   - This is the most common type of relationship.
   - **Example:**
     - Customer places Order.
     - Student enrolls in Course.

Relationship Degree (cont'd.)

3. **Ternary relationship (Degree 3)**
   - Three entities are associated.
   - **Example:**
     - Doctor prescribes a Drug to a Patient.
     - Supplier supplies a Product to a Store.

Relationship Degree (cont'd.)

FIGURE 7.16 The implementation of a ternary relationship

Database name: Ch07_Clinic

Table name: DRUG

DRUG_CODE	DRUG_NAME	DRUG_PRICE
AF15	Afgapan-15	25.00
AF25	Afgapan-25	35.00
DRO	Droalene Chloride	111.89
DRZ	Druzocholar Cryptolene	18.99
KO15	Koliabar Oxyhexalene	65.75
OLE	Oleander-Drizapan	123.95
TRYP	Tryptolac Heptadimetric	79.45

Table name: PATIENT

PAT_NUM	PAT_TITLE	PAT_LNAME	PAT_FNAME	PAT_INITIAL	PAT_DOB	PAT_AREACODE	PAT_PHONE
100	Mr.	Kolmycz	George	D	15-Jun-1942	615	324-5456
101	Ms.	Lewis	Rhonda	G	19-Mar-2005	615	324-4472
102	Mr.	Vandam	Rhett		14-Nov-1958	901	675-8993
103	Ms.	Jones	Anne	M	16-Oct-1974	615	898-3456
104	Mr.	Lange	John	P	08-Nov-1971	901	504-4430
105	Mr.	Williams	Robert	D	14-Mar-1975	615	890-3220
106	Mrs.	Smith	Jeanine	K	12-Feb-2003	615	324-7883
107	Mr.	Diante	Jorge	D	21-Aug-1974	615	890-4567
108	Mr.	Wiesenbach	Paul	R	14-Feb-1966	615	897-4358
109	Mr.	Smith	George	K	18-Jun-1961	901	504-3339
110	Mrs.	Genkazi	Leighla	W	19-May-1970	901	569-0093
111	Mr.	Washington	Rupert	E	03-Jan-1966	615	890-4925
112	Mr.	Johnson	Edward	E	14-May-1961	615	898-4387
113	Ms.	Smythe	Melanie	P	15-Sep-1970	615	324-9006
114	Ms.	Brandon	Marie	G	02-Nov-1932	901	882-0845
115	Mrs.	Saranda	Hermine	R	25-Jul-1972	615	324-5505
116	Mr.	Smith	George	A	08-Nov-1965	615	890-2984

Table name: DOCTOR

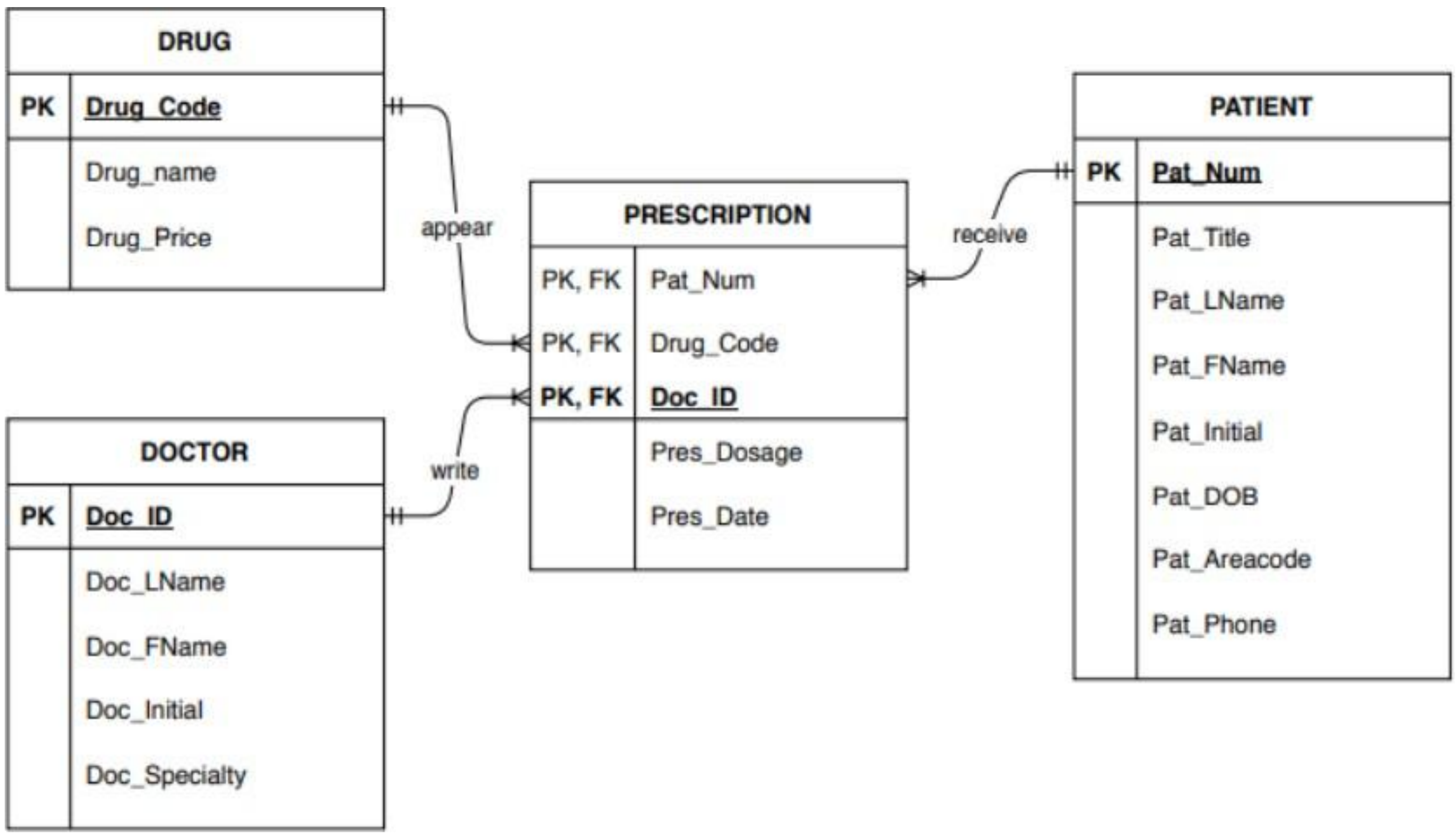
DOC_ID	DOC_LNAME	DOC_FNAME	DOC_INITIAL	DOC_SPECIALTY
29827	Sanchez	Julio	J	Dermatology
32445	Jorgensen	Annelise	G	Neurology
33456	Korenski	Anatoly	A	Urology
33989	LaGrange	George		Pediatrics
34409	Washington	Dennis	F	Orthopaedics
36221	McPherson	Katye	H	Dermatology
36712	Dreifag	Herman	G	Psychiatry
38995	Minh	Tran		Neurology
40004	Chin	Ming	D	Orthopaedics
40028	Feinstein	Denise	L	Gynecology

Table name: PRESCRIPTION

DOC_ID	PAT_NUM	DRUG_CODE	PRES_DOSAGE	PRES_DATE
32445	102	DRZ	2 tablets every four hours -- 50 tablets total	12-Nov-12
32445	113	OLE	1 teaspoon with each meal -- 250 ml total	14-Nov-12
34409	101	KO15	1 tablet every six hours -- 30 tablets total	14-Nov-12
36221	109	DRO	2 tablets with every meal -- 60 tablets total	14-Nov-12
38995	107	KO15	1 tablet every six hours -- 30 tablets total	14-Nov-12

SOURCE: Course Technology/Cengage Learning

Relationship Degree (cont'd.)

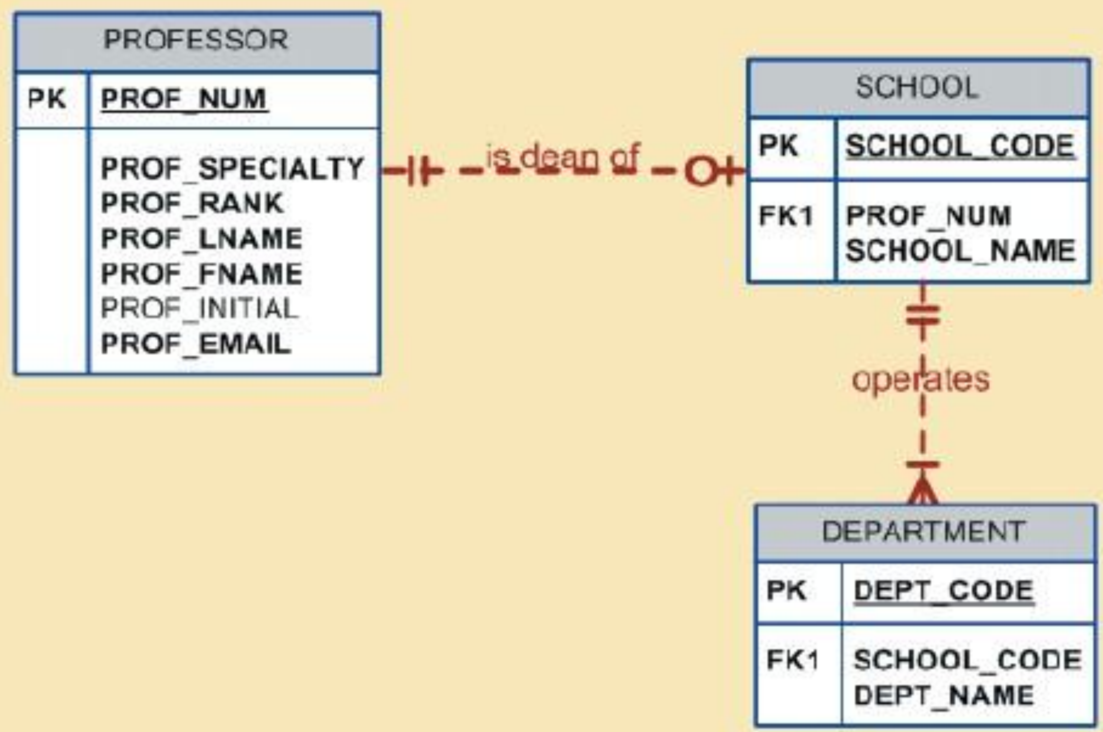


Developing an ER Diagram

- **Database design is an iterative process**
  - Create detailed narrative of organization's description of operations.
  - Identify business rules based on description of operations.
  - Identify main entities and relationships from business rules.
  - Develop initial ERD.
  - Identify attributes and primary keys that adequately describe entities.
  - Revise and review ERD.

Developing an ER Diagram (cont'd.)

FIGURE 7.26 The first Tiny College ERD segment



SOURCE: Course Technology/Cengage Learning

Developing an ER Diagram (cont'd.)

FIGURE 7.27 The second Tiny College ERD segment

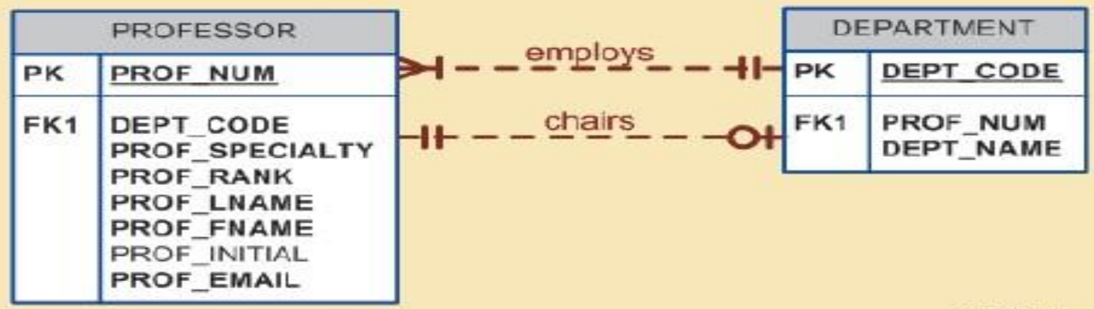
SOURCE: Course Technology/Cengage Learning

FIGURE 7.28 The third Tiny College ERD segment

SOURCE: Course Technology/Cengage Learning

Developing an ER Diagram (cont'd.)

FIGURE 7.29 The fourth Tiny College ERD segment

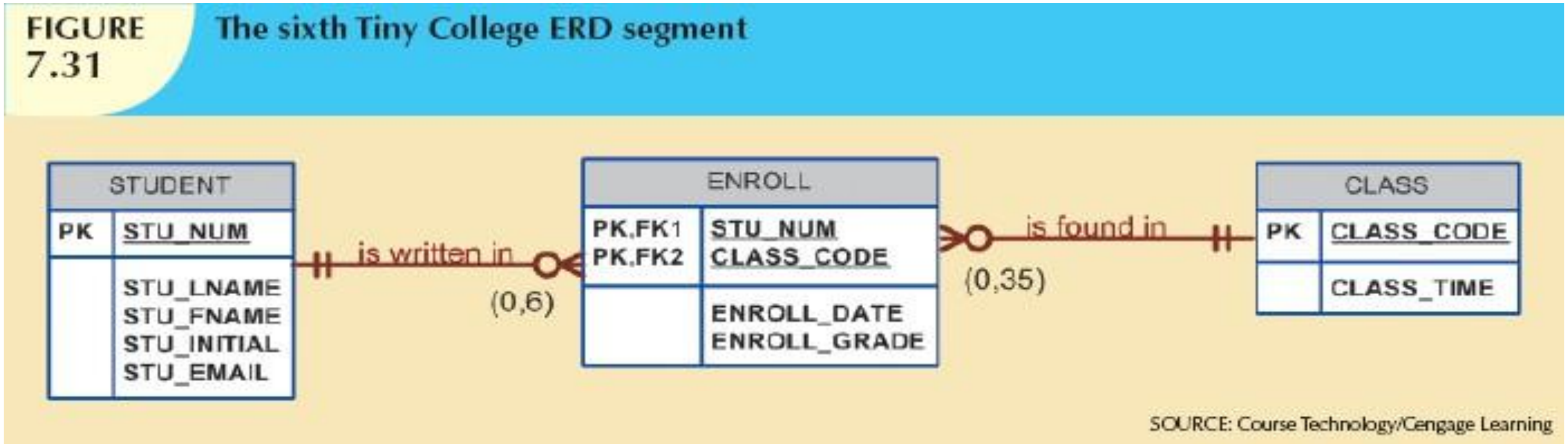


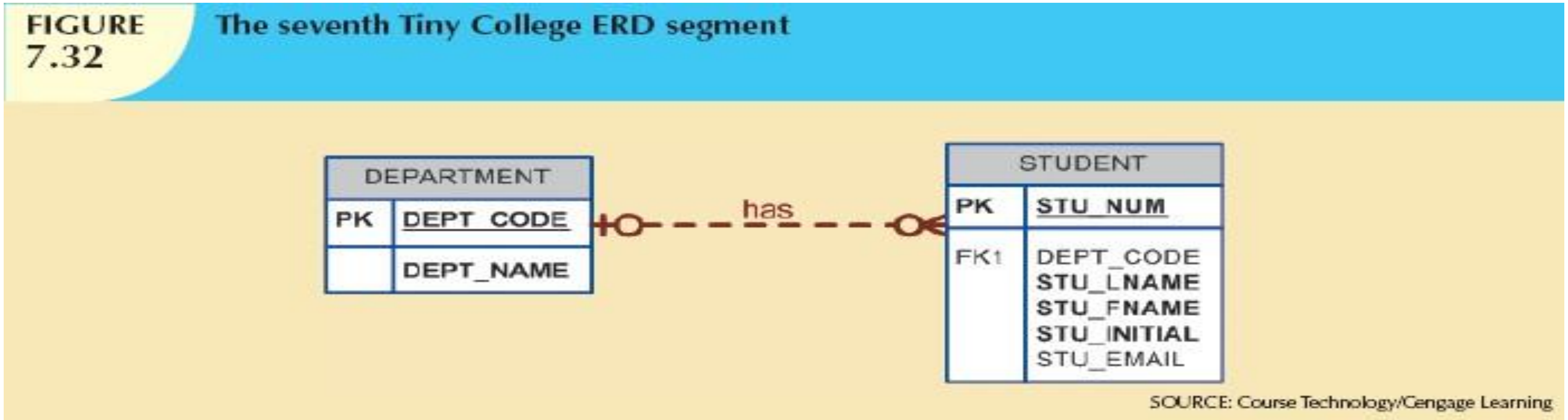
SOURCE: Course Technology/Cengage Learning

FIGURE 7.30 The fifth Tiny College ERD segment

SOURCE: Course Technology/Cengage Learning

Developing an ER Diagram (cont'd.)



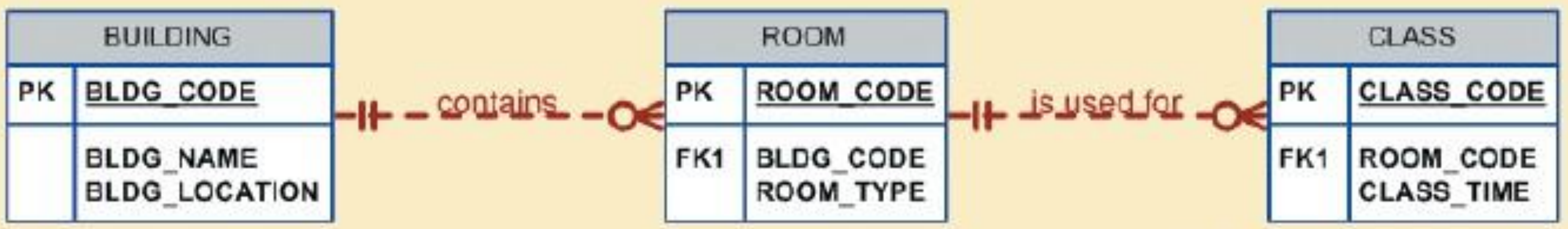


Developing an ER Diagram (cont'd.)

FIGURE 7.33 The eighth Tiny College ERD segment

SOURCE: Course Technology/Cengage Learning

FIGURE 7.34 The ninth Tiny College ERD segment



SOURCE: Course Technology/Cengage Learning

Developing an ER Diagram (cont'd.)

TABLE 7.4 Components of the ERM

ENTITY	RELATIONSHIP	CONNECTIVITY	ENTITY
SCHOOL	operates	1:M	DEPARTMENT
DEPARTMENT	has	1:M	STUDENT
DEPARTMENT	employs	1:M	PROFESSOR
DEPARTMENT	offers	1:M	COURSE
COURSE	generates	1:M	CLASS
PROFESSOR	is dean of	1:1	SCHOOL
PROFESSOR	chairs	1:1	DEPARTMENT
PROFESSOR	teaches	1:M	CLASS
PROFESSOR	advises	1:M	STUDENT
STUDENT	enrolls in	M:N	CLASS
BUILDING	contains	1:M	ROOM
ROOM	is used for	1:M	CLASS

Note: ENROLL is the composite entity that implements the M:N relationship "STUDENT enrolls in CLASS."

Summary

- **Entity relationship (ER) model**
  - Uses ERD to represent conceptual database as viewed by end user.
  - ERM's main components:
    - Entities
    - Relationships
    - Attributes
  - Includes connectivity and cardinality notations.
  - Connectivities and cardinalities are based on business rules.